Sound Pressure and Power

We're going to work through some of the conventions on sound pressure, sound power, and sound intensity. There's a lot of sloppiness in the definitions, so don't be surprised if what we see isn't 100% internally consistent. It'll be close enough for what we're doing. Sound pressure we can hear covers a very, very wide range, so it's natural to use a logarithmic scale. We've seen this before, but we'll go over it again in the context of power and pressure.

1. Decibels

The standard "unit" we'll use is the decibel or dB, which is defined to be 10 times the log base 10 of... something. If we want to take the log of something, it had better be dimensionless, so whenever we use a decibel, we *have* to have a reference value. Let's take acoustic intensity as an example. The acoustic intensity I is how much power is coming through an area, or Watts per square-metre. If we want to turn this into dB, we have to define a reference intensity. The standard is $I_0 = 10^{-12} W/m^2$. If you know how much power per area you have, then you can convert it into a standard dB measurement via:

$$L_I = 10 \log_{10}(I/I_0)\ dB \tag{1}$$

Where L_I is the level (L) of the sound intensity (I). If we have a microwatt per square metre (this is not a lot of power!) then we would find

$$L_I = 10 \log_{10}(10^{-6}/10^{-12} = 10 \log_{10}(10^6) = 60 dB \tag{2}$$

If increase that by a factor of 2, then the new intensity would be

$$L_I = 10 \log_{10}(2 * 10^{-6}/10^{-12} = 10 \log_{10}(2 * 10^6) = 63.01 dB \tag{3}$$

Doubling the intensity changed the level by 3.01 dB. If we doubled the intensity again, we would increase the level by another 3.01 dB, to 66.02 dB. If we increased the intensity by a factor of 10, the level would go up by $10 \log_1 010 = 10 dB$ since the log base 10 of 10 is 1.

People usually round 3.01 to 3, which gives us the following rules of thumb. If I increase my signal by a *factor* of 2, I increase the level by 3 dB. If I increase my signal by a factor of 10, I increase the level by 10 dB.

2. Sound Intensity and Power

As mentioned, we define sound intensity as the sound power per unit area. If we have a 1 watt speaker outside (so no reflections), then if we are 1 metre away, the power is coming through a sphere of radius 1m. The *area* of that sphere is $4\pi m^2$ (since $r = 1$), so the sound intensity is $\frac{1}{4\pi} W/m^2$. If we convert that to dB using the standard intensity reference value of $10^{-12} W/m^2$, we get:

$$L_I = 10 \log_{10}(\frac{1/4\pi}{1e-12}) = 10 \log_{10}(1e12/4\pi) = 109 dB \tag{4}$$

Sound power is the *total* amount of power a source is radiating. That is also often quoted in dB, and the reference level is $10^{-12}W$ or 1 picowatt. That is abbreviated L_{WA} and our 1-watt speaker would have

$$L_{WA} = 10\log_{10}(1/1e-12) = 120dB \tag{5}$$

Sound power is a property of the source, but sound intensity is how much power is reaching us at a given position. As we move around, the sound power from the source will remain constant, but the intensity will change. If there is no attenuation of the sound as it moves through the air, then the intensity will be the power divided by $4\pi r^2$. If we're at a distance r from our source that radiates with power P, then we'll have

$$L_I = 10\log_{10}(\frac{P/4\pi r^2}{1e-12} = 10\log_{10}(\frac{P}{1e-12}\frac{1}{4\pi r^2}) = 10\log_{10}(\frac{P}{1e-12}) + 10\log_{10}(\frac{1}{4\pi r^2}) \tag{6}$$

The first term is just the sound power in standard dB. The second term gives a $10\log_{10}(1/4\pi) = -10.99$ for the $1/4\pi$ part. The $log_{10}(r^{-2})$ is just $-2\log_{10}(r)$ since $\log(a^n) = n\log(a)$. That leaves us with the following conversion between sound power and sound intensity at a given distance r from the source (after rounding 10.99 to 11):

$$L_I = L_{WA} - 11 - 20\log_{10}(r/1m) \tag{7}$$

It's a bit silly that if you put a source with the reference power (so 0 dB) 1 metre away, you don't get the reference intensity, but pick up that factor of 11 dB, but I did not get to choose these values. The last term is called the *spreading loss*. As we go a factor of 10 further away, the intensity doesn't go down by a factor of 10 but instead a factor of 100, or 20 dB.

3. Sound Pressure Level

The sound pressure level or SPL is the RMS of the *change* in pressure from background. SPL is what our ears actually hear. Again, SPL is usually quoted in dB, and this time the reference level is 20 micropascal. This is about the threshold of human hearing. Because the energy transmitted by a wave goes like the pressure squared, the SPL in dB L_{PA} is defined as follows:

$$L_{PA} = 10\log_{10}(P^2/P_0^2) = 20\log_{10}(P/P_0) \tag{8}$$

where P^2 is the average of the pressure change squared, and P is the square root of that average.

It turns out that the intensity of a sound is the pressure squared times the acoustic impedance, which for air is $c_s\rho \simeq 410$ at STP. If you round that to 400, then the dB scale for intensity and pressure are the same, so $L_{PA} = L_I$. This is the scale you're most familiar with, and is what people use when they talk about how many dB a sound is. Note that the scales will diverge if you move away from air at STP since the conversion between pressure changes and energy depends on the background pressure. The reference levels also change depending on the medium. In particular, the reference pressure for sound in water is $1\mu Pa$, 20 times lower than in air.

4. Relations via Dimensional Analysis

I've quoted a lot of quantities here, and many of the formulas may have seemed to materialize out of thin air. It may not surprise you at this point that we can get rough answers from a little bit of physics insight and dimensional analysis.

First, what is the energy per volume in a sound wave? If you recall, the pressure is an energy density, so dimensionally it has to look like a pressure. However, if you'll recall from the very beginning of class, the work we have to do to squeeze half the air in a room goes like the distance we move *squared*. Basically, if I move a partition by some amount to the right, the fractional increase in pressure on the right side is proportional to the amount I moved and the decrease on the left is again proportional to the amount I moved. The force is proportional to how far I moved, and work is force times distance moved. Since the force is proportional to the distance, then the work must be proportional to the distance times distance, or distance squared. Since the pressure is proportional to the distance moved, then the work we do is proportional to the change in pressure squared. The work per unit volume must be proportional to the pressure change squared, but we know dimensionally it must be a pressure. Let's call the change in pressure P_{RMS}. The only other pressure we have floating around is the background pressure P_0, so the energy per volume must be the change in pressure squared divided by P_0. There might be a dimensionless factor floating around but we won't worry about that for now. We know that the energy per volume U must be:

$$U = \frac{P_{RMS}^2}{P_0} \tag{9}$$

because there isn't any other way to arrange the quantitites we have to give a pressure that is also proportional to P_{RMS}^2.

The energy flux, or sound intensity, has to be the energy density times the sound speed. If there is 1 Joule per cubic metre in sound energy, then I get 340 cubic metre's worth of energy going past me each second. The energy flux must then be

$$I = \frac{c_s P_{RMS}^2}{P_0} \tag{10}$$

Remember, we defined sound pressure level as

$$SPL = \left(\frac{P_{RMS}}{P_0}\right)^2 \tag{11}$$

To get the relation between the reference sound pressure level ($P_{ref} = 20$ micropascals) and intensity ($I_{ref} = 1$ picowatt per square metre) we use Equation **??** and set

$$I_{ref} = \frac{c_s P_{ref}^2}{P_0} \tag{12}$$

Recall that $c_s^2 = \gamma P_0/\rho_0$ so we can get rid of the background pressure (ignoring the γ) by replacing it with $c_s^2\rho_0$ and get

$$I_{ref} = \frac{P_{ref}^2}{c_s \rho_0} \tag{13}$$

There could have been other dimensionless factors floating around, but it turns out that Equation 13 is correct. The quantity on the bottom $c_s\rho_0$, the speed of sound times the density, is called the *acoustic impedance*. For air at STP, the impedance is 330 m/s times 1.22 kg per cubic metre, or 410 $kg/m^2/s$.

5. Acoustic Impedance

The unit of impedance $kg/m^2/s$ is called a *Rayl*. The acoustic impedance tells us how to convert the pressure (squared) in a sound wave P_{RMS} to the energy flow I. In air, the impedance will depend on temperature. The speed of sound scales like the square root of temperature, and density is inversely proprtional to temperature, so the impedance will be proportional to $T^{-1/2}$. The reference scales for acoustic intensity and SPL match when the impedance is 400 Rayl, so in general they will be slightly (but not a lot) off.

The impedance is also hugely important in understanding how sound travels between different media. For two different strings tied together, we saw the reflection was set by the wave speed. More generally, it is the impedance that sets how waves reflect/transmit at a junction between two materials. If we take our expressions from before for the reflected/transmited *power* of a wave, we swap in the impedance Z to get

$$I_R = I_A \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2} I_T = I_A \frac{4Z_1 Z_2}{(Z_1 + Z_2)^2} \tag{14}$$

These expressions are slightly different from the ones we derived earlier because there we worried about the *amplitude* of a wave. Now we're asking about the *intensity* or power carried by the wave. The power scales like the energy per length, which is proportional to the amplitude squared, but also picks up a *speed* of the wave, which is the transmitted intensity looks the way it does. As expected, if the impedances match, the reflected power is zero and the transmitted power is 100% of the incoming power.